Virtual Network Design and Implementation

Final Project Report

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11 July 2026

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Project Overview

This project builds a small virtual network in Cisco Packet Tracer and gets the core protocols running on it. The network has a router, a switch, two servers and three client PCs, plus one extra node standing in for the internet so NAT has somewhere to route to. Everything internal sits on 192.168.1.0/24.

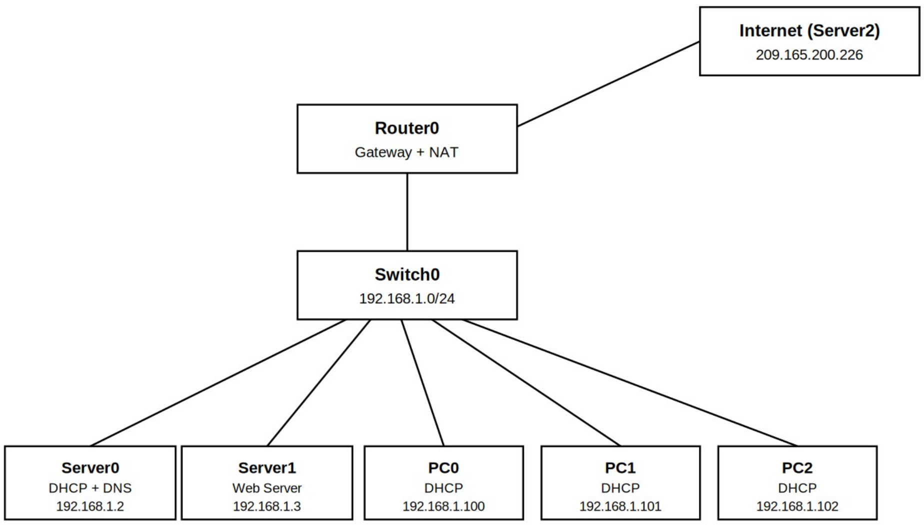
The point was to get it working and verified, not just drop devices onto a canvas and call it done. DHCP, DNS, HTTP, NAT, ARP, ICMP, TCP and UDP all had to be running and demonstrable by the end. Packet Tracer was used instead of separate virtual machines because everything runs inside one application, and its simulation mode lets you step through traffic at the packet level, which handles the packet analysis part without needing Wireshark on the side.

Objectives of the Project

The goal was to build a working network on 192.168.1.0/24 where clients get their addressing automatically from DHCP, resolve hostnames through DNS instead of typing raw IPs, load a page off a web server over HTTP, and reach an external network through NAT. Connectivity had to be tested at every stage with ping and traceroute before anything new was added on top, and TCP, UDP and ARP traffic had to be captured in simulation mode so the actual headers could be looked at.

Network Design Explanation

The internal network is 192.168.1.0/24. The router handles the default gateway on 192.168.1.1 and does NAT on its outside interface. The switch connects everything on the inside. Server0 sits at 192.168.1.2 and runs both DHCP and DNS. Server1 sits at 192.168.1.3 and runs the web server. The three PCs get their addresses from DHCP so they were never configured by hand. Server2 is on 209.165.200.226 on the other side of the router and stands in for the internet. A DNS record maps www.mynet.local to 192.168.1.3 so the clients can reach the web server by name.



Device	**IP Address**	**Subnet Mask**	**Gateway**	**Assigned**
Router0 (Gig0/0)	192.168.1.1	255.255.255.0	—	Static
Router0 (Gig0/1)	209.165.200.225	255.255.255.224	—	Static
Server0 (DHCP/DNS)	192.168.1.2	255.255.255.0	192.168.1.1	Static
Server1 (Web)	192.168.1.3	255.255.255.0	192.168.1.1	Static
PC0	192.168.1.100	255.255.255.0	192.168.1.1	DHCP
PC1	192.168.1.101	255.255.255.0	192.168.1.1	DHCP
PC2	192.168.1.102	255.255.255.0	192.168.1.1	DHCP
Server2 (external)	209.165.200.226	255.255.255.224	209.165.200.225	Static

Description of Networking Protocols Used

DHCP hands out IP addresses automatically so nothing has to be configured manually on the clients. It runs over UDP because the client has no IP yet when it sends the first message, so it can't do a TCP handshake. The client broadcasts from 0.0.0.0 and the server answers from port 67 to the client's port 68.

DNS turns a hostname like www.mynet.local into an IP address. It also runs over UDP. Without it every device would have to be reached by typing its raw IP.

ARP finds the MAC address that goes with an IP on the local network. A device broadcasts a request with the target MAC set to all zeros because it doesn't know it yet, and whoever owns that IP answers back with theirs.

ICMP is what ping and traceroute use. It doesn't carry real data, it just checks whether something is reachable and how many hops away it is.

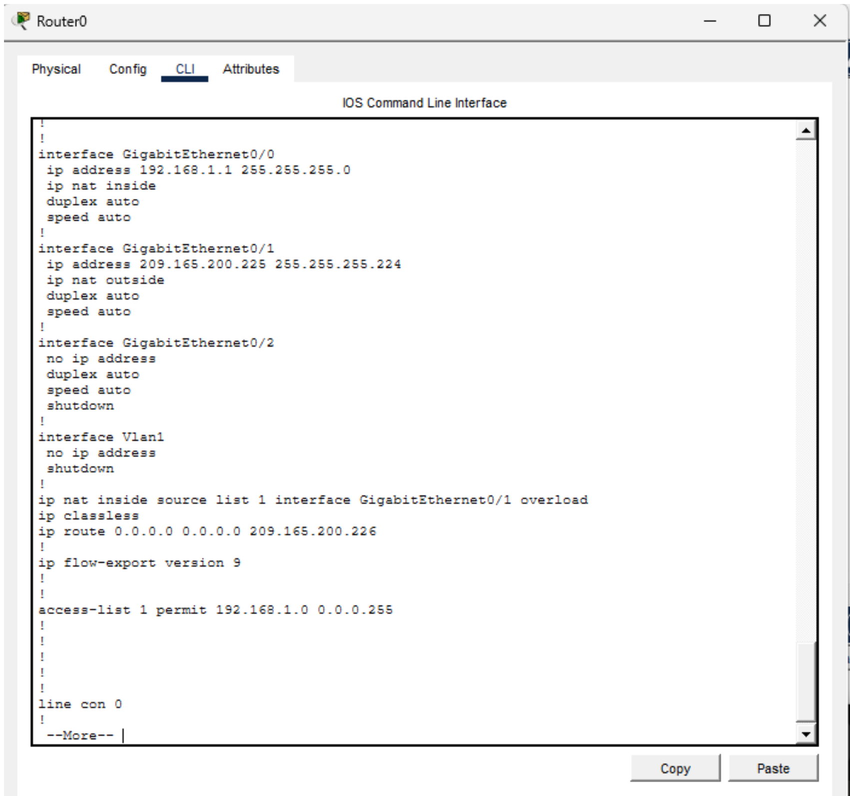
HTTP is the protocol the web server uses to send its page to the browser. It runs on top of TCP on port 80. TCP is connection oriented, so it sets up the connection first and makes sure everything arrives in order. UDP has no connection setup and just sends, which is why DHCP and DNS use it, they're short exchanges where the overhead of TCP isn't worth it.

NAT translates the private 192.168.1.x addresses to the router's outside address so internal devices can talk to an external network. Private addresses aren't routable on the outside, so without NAT nothing internal could leave.

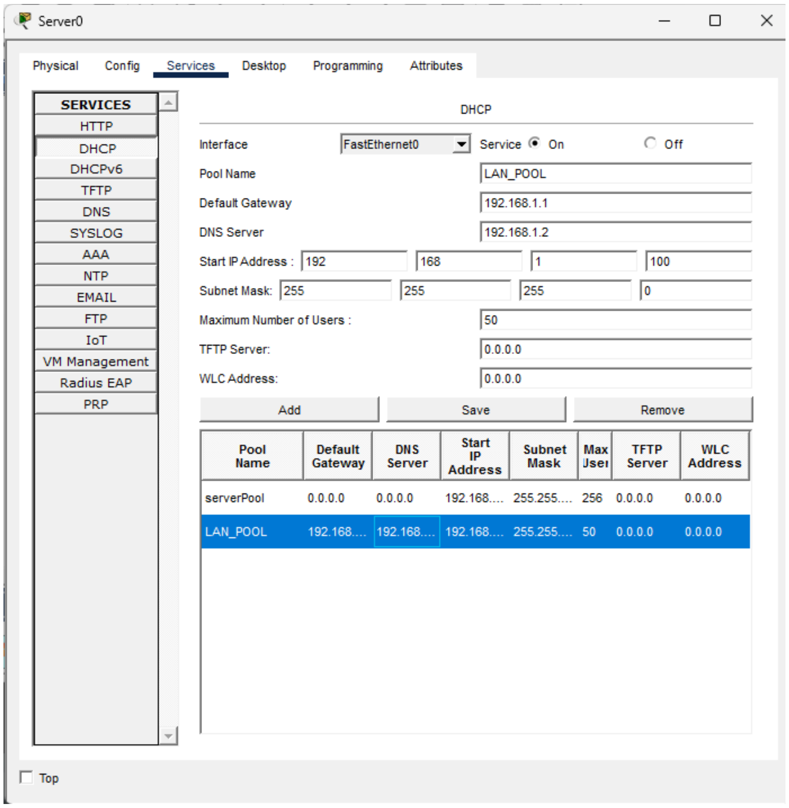
Device Configurations

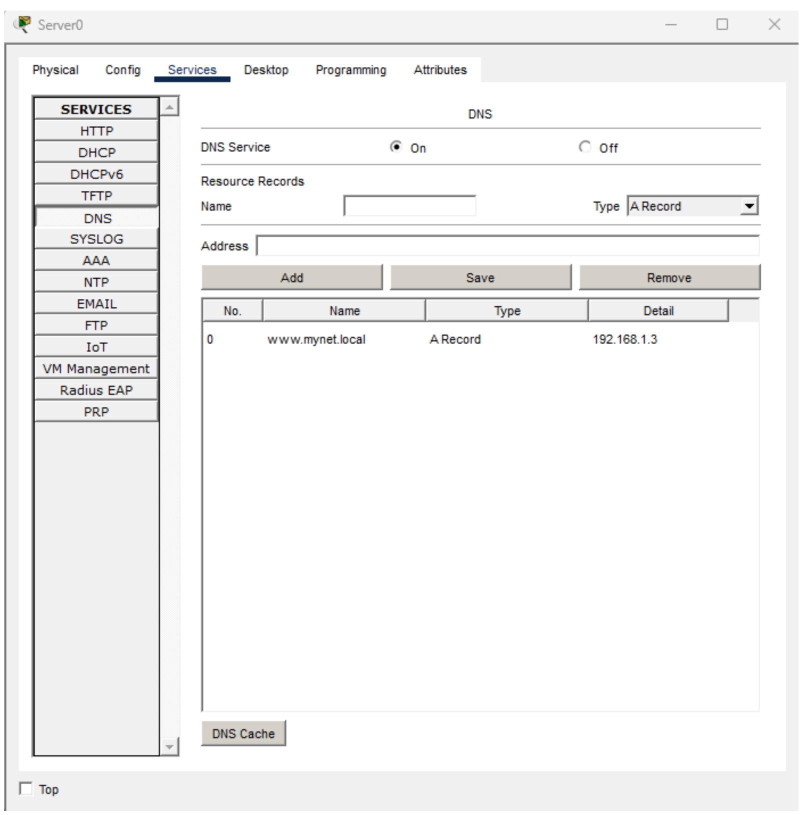
The router needed both interfaces addressed and turned on, since Cisco routers ship with every interface shut down. It also needed the NAT rules and a default route pointing at the external node.

```
interface g0/0
 ip address 192.168.1.1 255.255.255.0
 ip nat inside
 no shutdown
interface g0/1
 ip address 209.165.200.225 255.255.255.224
 ip nat outside
 no shutdown
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload
ip route 0.0.0.0 0.0.0.0 209.165.200.226
```



Server0 was given a static address of 192.168.1.2 and points its DNS at itself since it is the DNS server. The DHCP pool starts at 192.168.1.100 and hands out the gateway and DNS server along with the address. The DNS record maps www.mynet.local to the web server.


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
LAN_POOL
Default Gateway
192.168.1.1
DNS Server
192.168.1.2
Start IP Address :
192
168
1
100
Subnet Mask:
255
255
255
0
Maximum Number of Users :
50
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool 0.0.0.0 0.0.0.0 192.168.... 255.255.... 256 0.0.0.0 0.0.0.0
LAN_POOL 192.168.... 192.168.... 192.168.... 255.255.... 50 0.0.0.0 0.0.0.0
Top


Server0
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SERVICES
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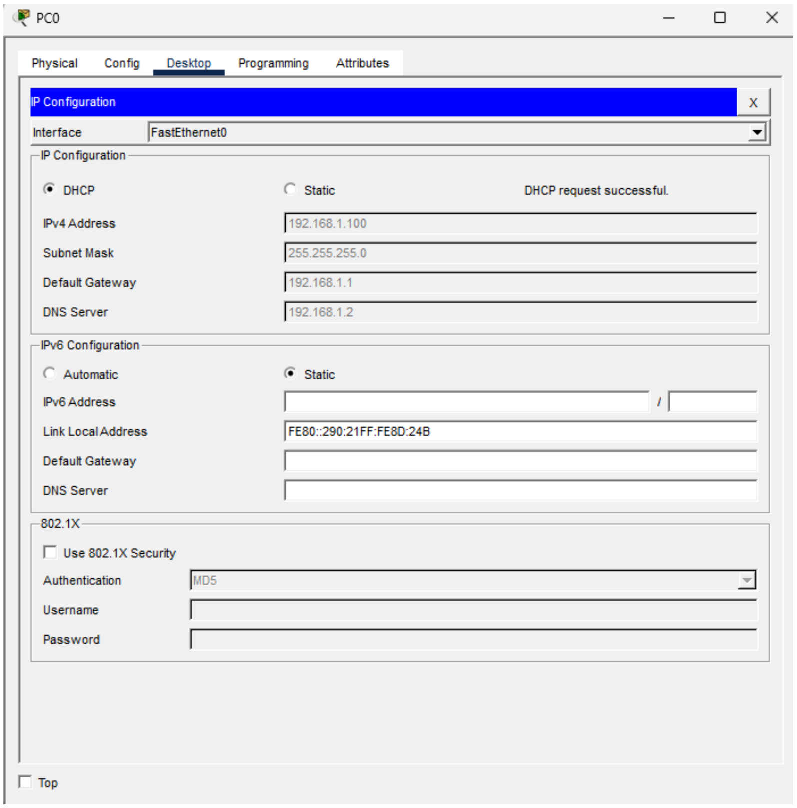
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0 www.mynet.local A Record 192.168.1.3
DNS Cache
Top

Server1 was given 192.168.1.3 with HTTP and HTTPS both switched on. The three PCs were set to DHCP and nothing else was typed in on them.



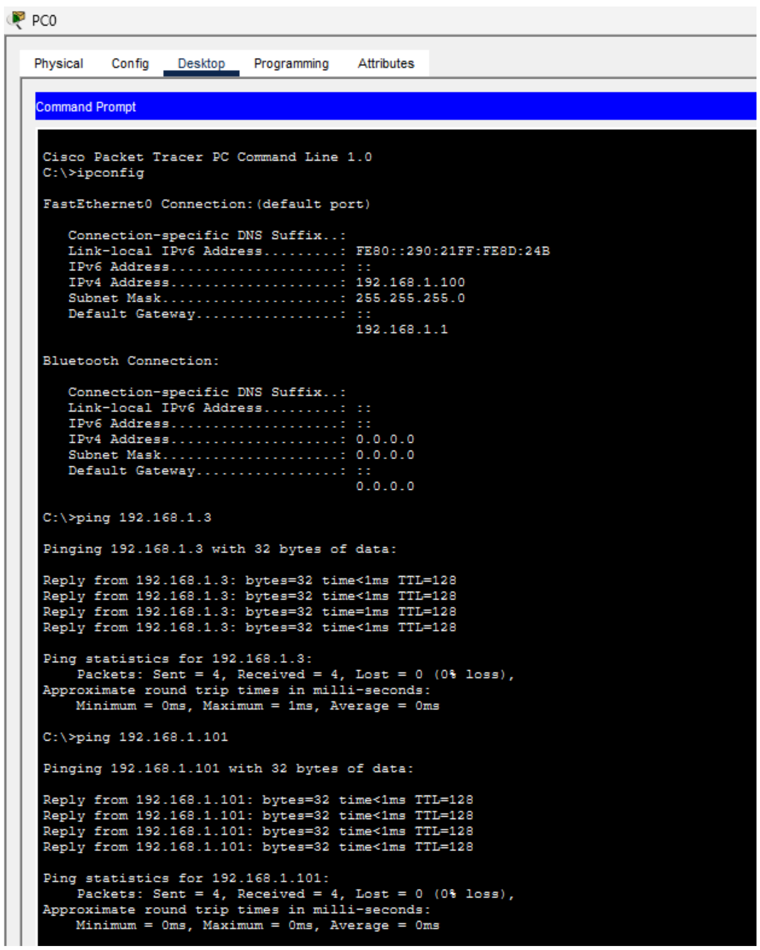
Network Testing and Results

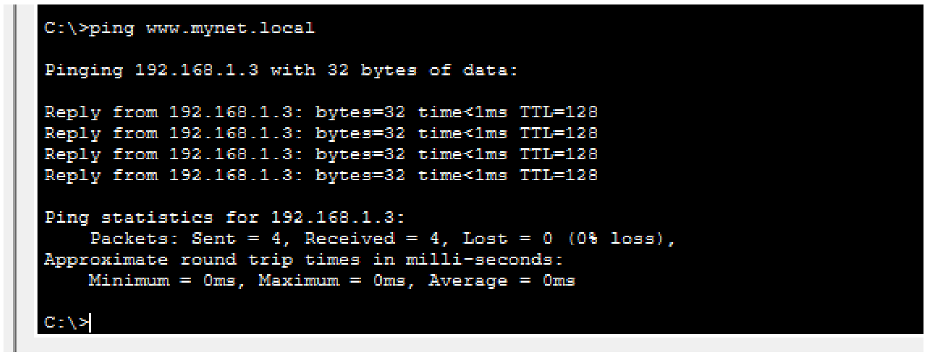
Testing was done stage by stage. Addressing first, then pings, then the services on top of that. Doing it that way meant if something broke it was obvious what caused it, instead of configuring everything at once and then trying to debug a network that had never worked.

Test	Protocol	Result
PCs set to DHCP	DHCP	All three got .100, .101, .102 with nothing typed in
ping 192.168.1.3 from PC0	ICMP	4 replies, 0% loss
ping 192.168.1.101 from PC0	ICMP	4 replies, 0% loss
ping www.mynet.local	DNS	Resolved to 192.168.1.3, 4 replies, 0% loss
ping 209.165.200.225 from Server2	ICMP	4 replies, 0% loss
tracert 209.165.200.225	ICMP	Reached the router in one hop
http://www.mynet.local in browser	HTTP/TCP	Page loaded by hostname
show ip nat statistics	NAT	Inside and outside interfaces listed correctly
Simulation capture, page load	TCP	Ports 1029 to 80, seq and ack numbers visible

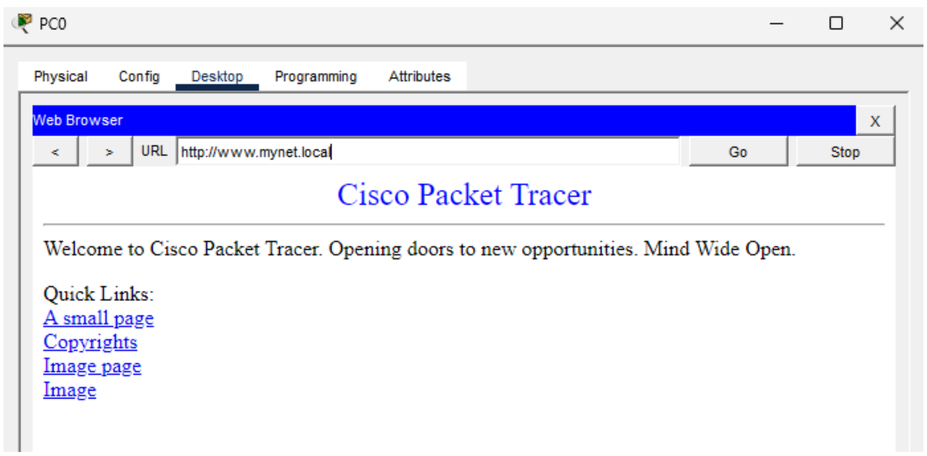
Simulation capture, DHCP	UDP	Ports 68 to 67, broadcast from 0.0.0.0
Simulation capture, first ping	ARP	Request with target MAC all zeros
ping 209.165.200.226 from PC0	ICMP/NAT	Timed out, see Challenges

PC0 shows the address it pulled from DHCP and pings both the web server and PC1 with no loss. Pinging the hostname instead of the IP also works and resolves to 192.168.1.3, which means the client got the DNS server from DHCP, asked Server0, and Server0 answered with the right address.

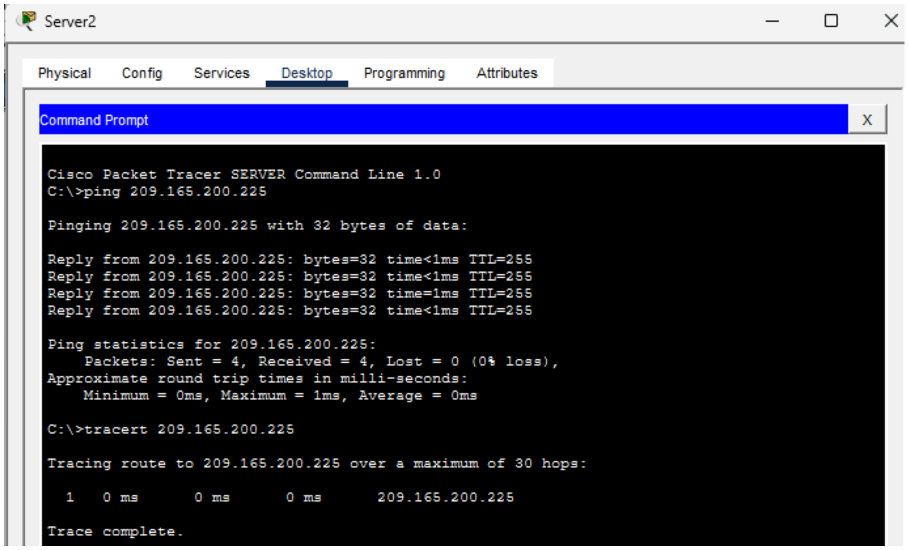




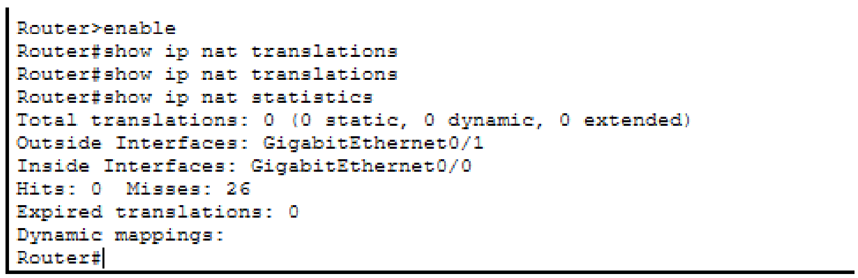
The browser on PC0 loads the page by typing the hostname rather than the IP. That one screen proves DNS, HTTP and TCP are all working at the same time.



On the external side, Server2 reaches the router's outside interface with no loss and traceroute gets there in one hop, so the outside link is up and routing. The NAT statistics show Gig0/1 as the outside interface and Gig0/0 as the inside, which matches the configuration.

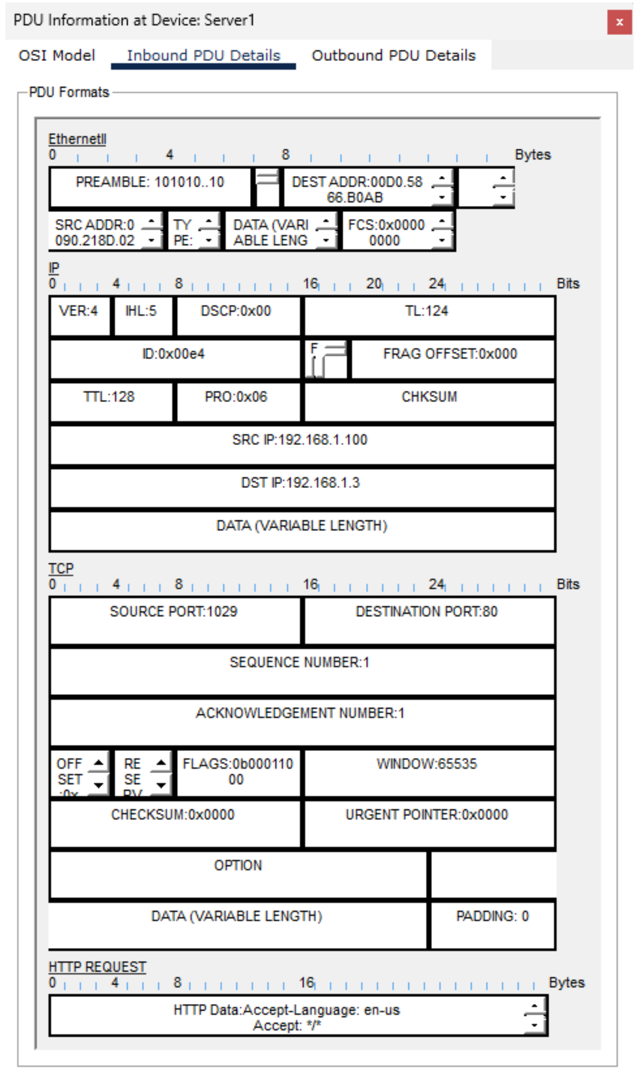


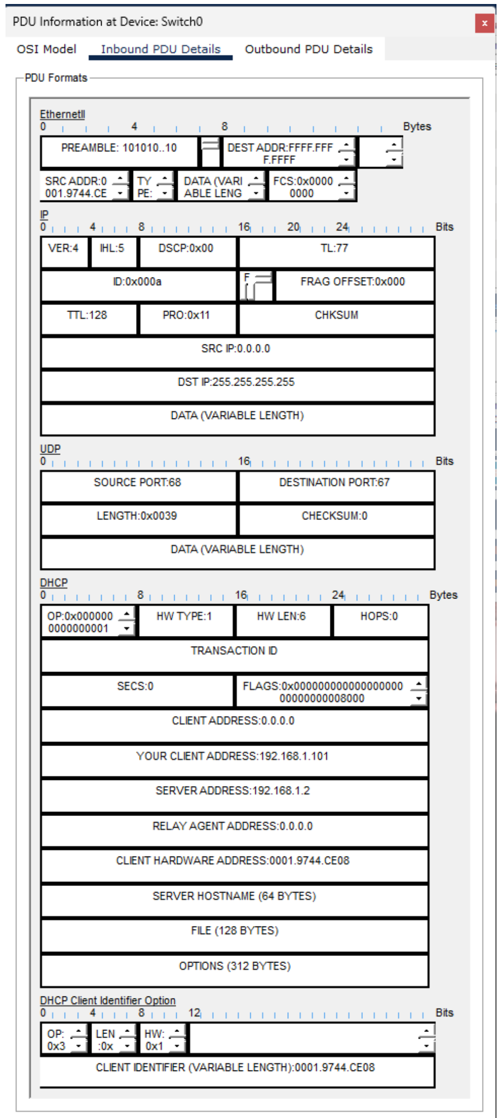

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 209.165.200.225

Pinging 209.165.200.225 with 32 bytes of data:

Reply from 209.165.200.225: bytes=32 time<1ms TTL=255
Reply from 209.165.200.225: bytes=32 time<1ms TTL=255
Reply from 209.165.200.225: bytes=32 time=1ms TTL=255
Reply from 209.165.200.225: bytes=32 time<1ms TTL=255

Ping statistics for 209.165.200.225:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 209.165.200.225

Tracing route to 209.165.200.225 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      209.165.200.225

Trace complete.
```

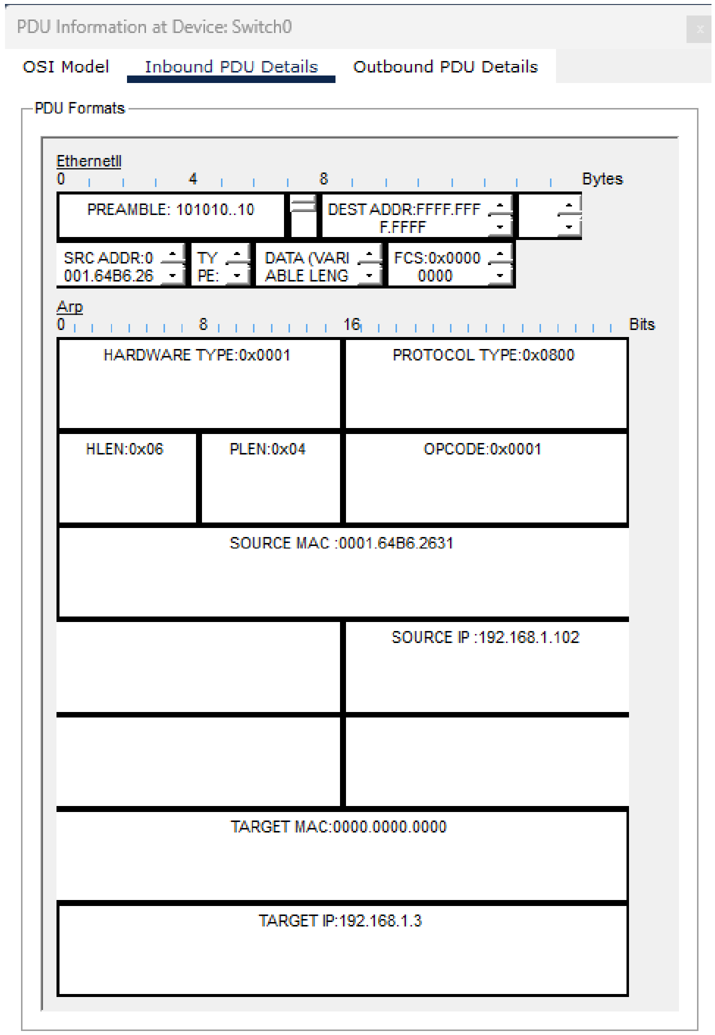


```
Router>enable
Router#show ip nat translations
Router#show ip nat translations
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 0  Misses: 26
Expired translations: 0
Dynamic mappings:
Router#
```

Simulation mode was used to capture traffic and open the packets up. The TCP capture of the web request shows source port 1029 going to destination port 80 with the sequence and acknowledgement numbers and the flags. The DHCP capture shows ports 68 to 67 inside a UDP header, with the source IP as 0.0.0.0 broadcasting to 255.255.255.255 because the client has no address yet, and 192.168.1.101 being offered to PC1 underneath. The ARP capture shows PC2 asking who has 192.168.1.3, with the target MAC left as all zeros because that is the exact thing it is trying to find out.


PDU Information at Device: Server1
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
EthernetII
Bytes
PREAMBLE: 101010..10
DEST ADDR:00D0.58 66.B0AB
SRC ADDR:0 090.218D.02
TY PE:
DATA (VARI ABLE LENG
FCS:0x0000 0000
IP
Bits
VER:4
IHL:5
DSCP:0x00
TL:124
ID:0x00e4
FRAG OFFSET:0x000
TTL:128
PRO:0x06
CHKSUM
SRC IP:192.168.1.100
DST IP:192.168.1.3
DATA (VARIABLE LENGTH)
TCP
Bits
SOURCE PORT:1029
DESTINATION PORT:80
SEQUENCE NUMBER:1
ACKNOWLEDGEMENT NUMBER:1
OFF SET
RE SE
FLAGS:0b000110 00
WINDOW:65535
CHECKSUM:0x0000
URGENT POINTER:0x0000
OPTION
DATA (VARIABLE LENGTH)
PADDING: 0
HTTP REQUEST
Bytes
HTTP Data:Accept-Language: en-us
Accept: */*


PDU Information at Device: Switch0
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
EthernetII
PREAMBLE: 101010..10
DEST ADDR:FFFF.FFFF.FFFF
SRC ADDR:0001.9744.CE
TYPE:
DATA (VARIABLE LENG
FCS:0x00000000
IP
VER:4
IHL:5
DSCP:0x00
TL:77
ID:0x000a
FRAG OFFSET:0x000
TTL:128
PRO:0x11
CHKSUM
SRC IP:0.0.0.0
DST IP:255.255.255.255
DATA (VARIABLE LENGTH)
UDP
SOURCE PORT:68
DESTINATION PORT:67
LENGTH:0x0039
CHECKSUM:0
DATA (VARIABLE LENGTH)
DHCP
OP:0x0000000000000001
HW TYPE:1
HW LEN:6
HOPS:0
TRANSACTION ID
SECS:0
FLAGS:0x0000000000000000000000000008000
CLIENT ADDRESS:0.0.0.0
YOUR CLIENT ADDRESS:192.168.1.101
SERVER ADDRESS:192.168.1.2
RELAY AGENT ADDRESS:0.0.0.0
CLIENT HARDWARE ADDRESS:0001.9744.CE08
SERVER HOSTNAME (64 BYTES)
FILE (128 BYTES)
OPTIONS (312 BYTES)
DHCP Client Identifier Option
OP: 0x3
LEN:0x
HW: 0x1
CLIENT IDENTIFIER (VARIABLE LENGTH):0001.9744.CE08



Challenges Faced and Solutions

The first problem was the router interfaces. Both came up dead even though the cables were connected and the switch side was green. Cisco routers ship with every interface administratively shut down, so the link stays red until you turn it on. Adding no shutdown to both interfaces fixed it and the lights went green straight away.

The second one wasted more time than it should have. The ip address command kept getting rejected with an invalid input error. The IP and the subnet mask have to be separated by a space or the router reads them as one long string and doesn't know what to do with it.

The third one took the longest. Pinging the external node from PC0 timed out every time, and tracert did the same for all 30 hops. The NAT configuration was checked line by line against the running config and it was correct, the inside and outside interfaces were assigned properly and the access list matched the internal subnet. Pinging from Server2 to the router's outside interface worked fine with no loss, which meant the outside link was healthy and the problem was on the return path. Packet Tracer doesn't handle ICMP replies coming back through NAT reliably, so the ping goes out and never makes it back. Rather than burn more time on it, routing was verified from the other direction using Server2 and the NAT setup was documented with show ip nat statistics and the running config.

The NAT translation table also came back empty at first. The browser test goes from PC0 to Server1 and both of those are on the same internal subnet, so that traffic never crosses NAT at all, it just goes through the switch. Nothing was wrong with the config, it just wasn't generating the kind of traffic that produces a translation entry.

Conclusion

The network works. Clients boot and get everything they need from DHCP without anything being typed in, they resolve a hostname through DNS, they load a page off the web server over HTTP, and the router routes between the internal network and the external one. All of it was verified with ping, traceroute, the browser and packet captures rather than just assumed.

Building it one stage at a time was the thing that made it manageable. Every time something broke it broke right after the one change that caused it, so there was never much guessing involved. Seeing the packets opened up in simulation mode also made the protocols click in a way that reading about them doesn't. The DHCP packet broadcasting from 0.0.0.0 because the client has no address yet, and the ARP request with a target MAC of all zeros because that is the thing it is asking about, both make a lot more sense once you can see the fields sitting there in the header.